

Dual-Ligand System for Mild Decarbonylative Suzuki–Miyaura Cross-Coupling of Aryl Chlorides

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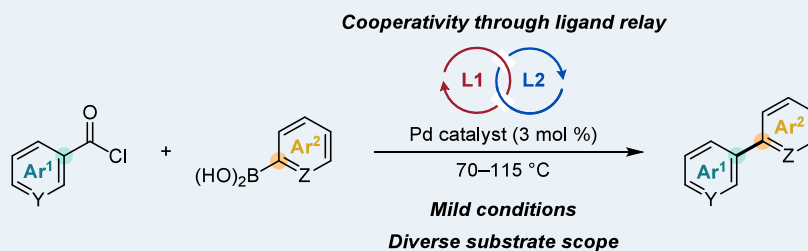
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ABSTRACT: Cooperativity between a pair of phosphine ligands enables general Pd-catalyzed decarbonylative Suzuki–Miyaura cross-couplings between (hetero)aryl chlorides and (hetero)arylboronic acids under mild conditions. Experimental and computational studies support a ligand-relay mechanism in which each phosphine preferentially promotes different elementary steps, enhancing the yield and selectivity relative to using either ligand alone. These results validate empirical, mechanism-agnostic screening through pooling–deconvolution as a means for identifying synthetically enabling catalytic methods and mechanisms for multiligand cooperativity.

KEYWORDS: Cooperative catalysis, Pd catalysis, Cross-coupling, Ligand relay, Decarbonylation

INTRODUCTION

Arylcarboxylic acids and their derivatives are appealing substrates for aryl–aryl cross-coupling reactions due to their diverse abundance and direct accessibility from feedstock chemicals.^{1–3} Arylcarboxylic acids have been engaged as nucleophiles in catalytic decarboxylative cross-coupling reactions with aryl halides, as pioneered by Gooßen and co-workers.^{4,5} While these methodologies enable direct engagement of the parent carboxylic acids, they require forcing reaction temperatures (>160 °C) and often specific substitution patterns to overcome the tremendous kinetic barrier associated with decarboxylation and transmetalation.⁶ Alternatively, carboxylic acid derivatives can participate as electrophiles in decarbonylative cross-couplings with arylboronic acids, which affords access to chemical space uncharted by decarboxylative acid–halide couplings.⁷ Certain carboxylic acid derivatives have been shown to participate in high-yielding and general decarbonylative cross-couplings,^{8–14} leveraging the ability of metals such as Ni, Rh, and Pd to promote extrusion of CO from a metal–acyl complex.^{15–19}

Among carboxylic acid derivatives, acyl chlorides are the most reliably accessible class for use in pharmaceutical process chemistry.²⁰ While their application in acyl Suzuki cross-couplings is well-developed,²¹ they remain difficult to engage directly as substrates in decarbonylative couplings.^{22–28} Only two catalytic methods have been described that employ them in Pd-catalyzed decarbonylative cross-coupling toward biar-

yls.^{29,30} These approaches require either discrete decarbonylation to the corresponding aryl chlorides followed by a separate coupling step (Figure 1A, 2)²⁹ or high catalyst loading and temperature (Figure 1A, 1).³⁰ These limitations contrast sharply with the generality of modern methods for cross-coupling with aryl halides,³¹ reflecting not only the well-documented kinetic challenge of decarbonylation^{32,33} but also the fundamental difficulty of requiring a single ancillary ligand to support a diverse set of elementary steps (oxidative addition, decarbonylation, transmetalation, and reductive elimination).³⁰

We posited that a multiligand approach could enable efficient catalysis of decarbonylative couplings with aryl chlorides under mild conditions, by distributing the requisite steps over metal complexes supported by different ligands.^{34–39} With the Pd-catalyzed decarbonylative cross-coupling between 1-naphthoyl chloride **1d** and 4-fluorophenylboronic acid **2b** as the model reaction, we reported recently an empirical discovery effort based on combinatorial pooling and deconvolution to accelerate the discovery of ligand cooperativity (Figure 1B).⁴⁰ This campaign resulted in the

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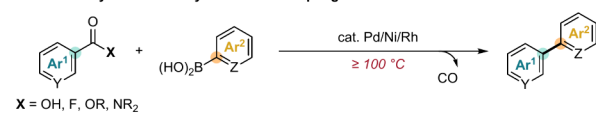
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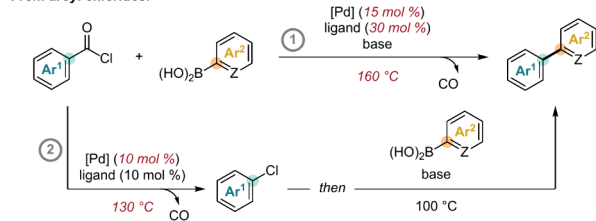
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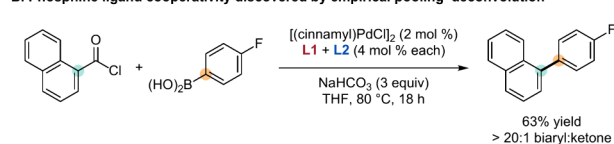
A. Metal-catalyzed decarbonylative cross-couplings



From aryl chlorides:



B. Phosphine ligand cooperativity discovered by empirical pooling–deconvolution



C. This work: ligand cooperativity enables efficient and general reaction

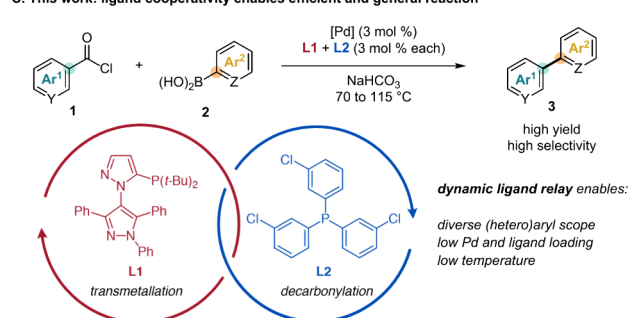


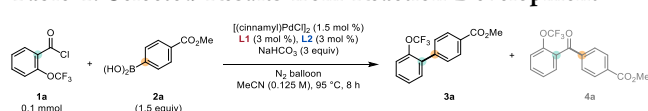
Figure 1. (A) State of the art in metal-catalyzed decarbonylative cross-coupling. (B) Discovery of cooperativity between phosphine ligands from the pooling–deconvolution campaign. (C) An efficient and general decarbonylative cross-coupling platform powered by ligand cooperativity.

identification of several pairs of phosphorus-centered ligands exhibiting evidence of catalytic cooperativity, whereby the reaction yield with both ligands employed exceeded the maximum yield of each ligand alone. Among these, we identified the combination of BippyPhos **L1** and tris(3-chlorophenyl)phosphine **L2** as a particularly appealing pair for further evaluation, given that both ligands are inexpensive⁴¹ and in initial experiments promote the model reaction with yields comparable to the state of the art but at drastically lower metal/ligand loadings and temperature (Figure 1B).

This initial evidence of cooperativity presented an ideal starting point to develop a general Pd-catalyzed decarbonylative cross-coupling between aryl chlorides and arylboronic acids as well as to elucidate potentially transferable mechanistic principles underlying cooperativity. Here, we describe a cross-coupling protocol that is compatible with diverse aryls and heteroaryls on both electrophilic and nucleophilic partners and that requires lower temperatures and catalyst loadings relative to previous reports. Experimental and computational investigations reveal a mechanism for cooperativity originating from dynamic ligand relay.⁴²

To ensure the generality of the catalytic system, reaction optimization was performed on **1a** and **2a**, a substrate pair different from the previous target reaction (Table 1). This effort identified acetonitrile as a superior solvent to

Table 1. Selected Results from Reaction Development



entry	variation	yield of 3a (%) ^a	yield of 4a (%) ^a	3a/4a
1	none	77	1	>20:1
2	THF	18	1	18:1
3	2.0 equiv NaHCO ₃	68	0	>20:1
4	1.0 equiv 2a	52	0	>20:1
5	K ₂ CO ₃ (3 equiv)	9	11	0.86
6	K ₃ PO ₄ (3 equiv)	18	13	1.33
7	sealed, under air	45	2	>20:1
8	115 °C, sealed	81	1	>20:1
9	L1 only (3 mol %)	28	2	14:1
10	L1 only (6 mol %)	23	1	>20:1
11	L2 only (3 mol %)	37	1	>20:1
12	L2 only (6 mol %)	46	1	>20:1

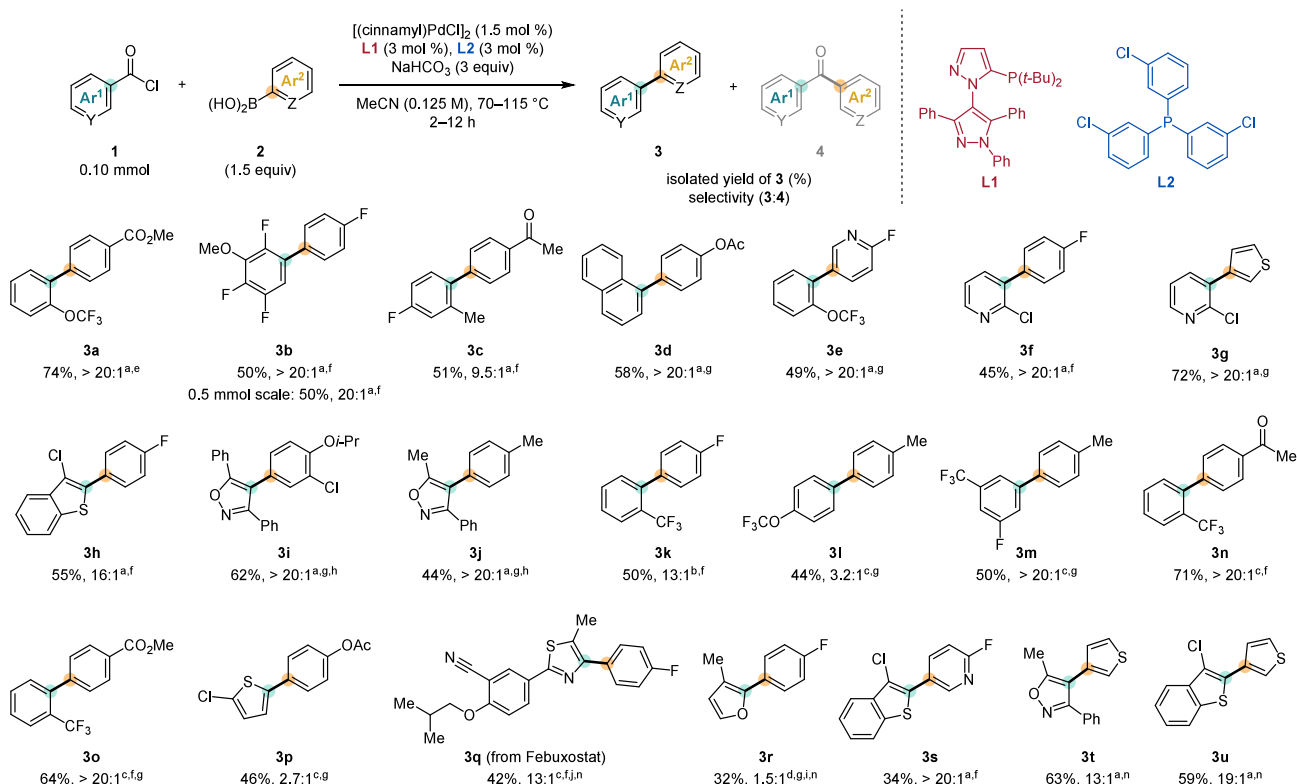
^aYields were determined by GC analysis and calibrated against authentic product standards.

tetrahydrofuran (entry 2), enabling high yield (77%) and excellent selectivity (>20:1) at even lower Pd and ligand loadings (3 mol % Pd and 6 mol % P ligand), as well as higher concentration (0.125 M) at 95 °C (entry 1).

Reducing or eliminating the excess of NaHCO₃ or boronic acid **2a** employed in the reaction led to modest yet appreciable decreases in yield (entries 3 and 4). Substituting NaHCO₃ with commonly used bases in Suzuki–Miyaura reactions such as K₂CO₃ (entry 5) and K₃PO₄ (entry 6) proved deleterious to both selectivity and yield. We propose that stronger bases may favor transmetalation at the Pd(acyl) stage prior to decarbonylation (*vide infra*), resulting in a pathway to formation of undesired biaryl ketone **4**. Performing the reaction in a sealed vial rather than with an attached nitrogen-filled balloon led to a decrease in yield to 45% (entry 7). This observation is consistent with reversibility of the decarbonylation step and increased headspace with the balloon shifting the Pd(acyl) vs Pd(aryl) equilibrium toward decarbonylation by reducing CO partial pressure. However, increasing the temperature to 115 °C under the sealed-vial conditions reverses the drop in yield (entry 8). Using **L1** or **L2** alone at either 3 or 6 mol % loading resulted in markedly lower yields and selectivities (entries 9–12), confirming ligand cooperativity under these optimized conditions.

Under the developed reaction conditions, (hetero)aryl chlorides derived from a diverse array of aryl and heteroaryl carboxylic acids were found to be suitable coupling partners (Figure 2A). This system enables efficient and selective coupling of a variety of such substrates including pyridines **3f** and **3g**, benzothiophene **3h**, **3s**, and **3u**, isoxazoles **3i**, **3j**, and **3t**, and thiazole **3q**. Four examples in the substrate scope (**3g**, **3s**, **3t**, and **3u**) feature heterocycles on both coupling partners. Electron-rich 2-thiophenecarbonyl chloride (**3p**) and 2-furoyl chloride (**3r**) proved to be less reactive, requiring higher temperature and affording lower biaryl-to-ketone selectivity but still providing cross-coupled product in preparatively useful yields. The boronic acid scope includes various aryl and heteroaryl scaffolds; both coupling partners also feature a wide range of electron-withdrawing and electron-donating substituents. Compatible functional groups include esters (**3a**, **3d**, **3o**, **3p**), trifluoromethoxy groups (**3a**, **3e**, **3l**),

A. Substrate scope



B. Comparison of yield and selectivity with dual-ligand and single-ligand systems

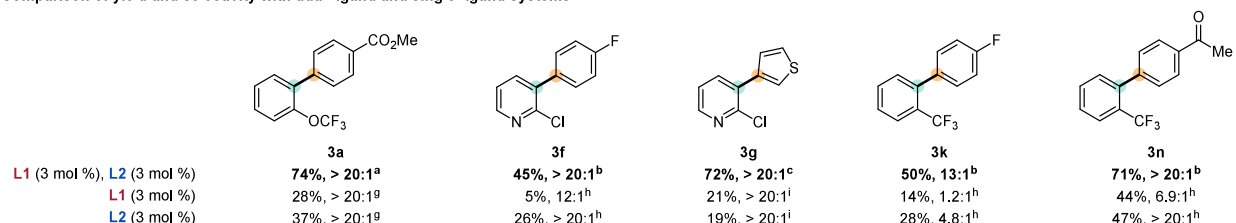


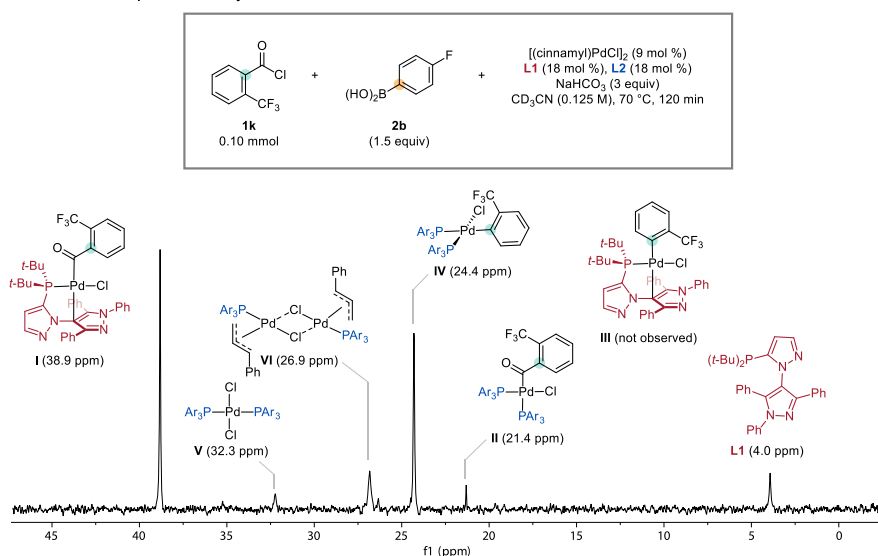
Figure 2. (A) Substrate scope. ^a90 °C, 12 h. ^b70 °C, 12 h. ^c105 °C, 8 h. ^d115 °C, 2 h. ^eSelectivity was determined by a GC-FID analysis of the crude reaction mixture. ^fSelectivity was determined by quantitative ¹⁹F NMR analysis of the crude reaction mixture. ^gSelectivity was determined by quantitative ¹H NMR analysis of the crude reaction mixture. ^hSelectivity was determined by GC-MS analysis of the crude reaction mixture. ⁱ Na_2CO_3 was used instead of NaHCO_3 . ^j5 mol % each of $[(\text{cinnamyl})\text{PdCl}]_2$, **L1**, and **L3** were used instead of 3 mol % each. ^kYield and selectivity were determined by GC-FID analysis of the crude reaction mixture. ^lYield and selectivity were determined by quantitative ¹⁹F NMR analysis of the crude reaction mixture. ^mYield and selectivity were determined by quantitative ¹H NMR analysis of the crude reaction mixture. ⁿ(3,5- $\text{CF}_3\text{C}_6\text{H}_3$)₃Ph (**L3**) was used instead of **L2**. (B) Assessment of cooperative effects on the yield and selectivity for randomly selected substrate pairs.

ketones (**3c**, **3n**), ethers (**3b**, **3i**, **3q**), nitriles (**3q**), and (hetero)aryl chlorides (**3f**, **3g**, **3h**, **3i**, **3p**), which are common electrophiles in other cross-coupling and nucleophilic aromatic substitution reactions. This functional-group tolerance renders the new decarbonylative method well suited for subsequent synthetic elaborations. The decarbonylative cross-coupling of Febuxostat-derived aryl chloride proceeded with useful yields and good selectivity to form **3q** when electron-deficient **L3** was used in place of **L2**. Aryl chlorides lacking *ortho*-substituents required higher temperatures and afforded reduced selectivity for the biaryl product in some cases (**3m**, **3q**, and **3l**).

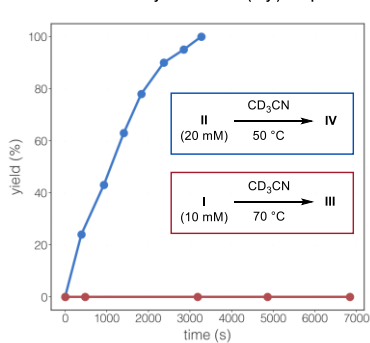
Five pairs of reaction partners were selected at random to gain broader insight into the contribution of ligand cooperativity to yield and selectivity (Figure 2B). Compared with **L1** or **L2** alone, the dual-ligand system afforded markedly higher reaction yields for all five products and higher selectivity

for three of the five, underscoring the generality of the ligand cooperativity phenomenon in this system.

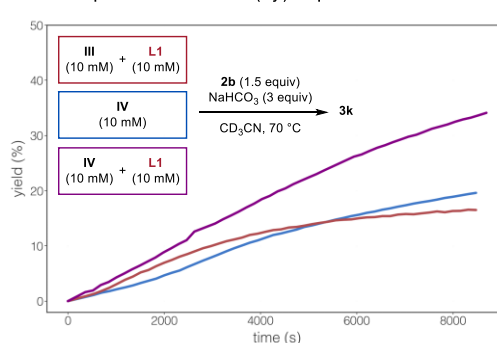
We sought to elucidate the mechanistic origins of ligand cooperativity through an in-depth study of the cross-coupling between aryl chloride **1k** and boronic acid **2b** to form biaryl **3k**, which proceeds readily at 70 °C in MeCN and can be monitored conveniently *in situ*. To understand the basis for cooperativity, we aimed to determine the role of each ligand in decarbonylation, transmetalation (TM), and reductive elimination (RE). First, given the generally accepted mechanism for decarbonylative cross-coupling, we considered complexes (**L1**)Pd(acyl)(Cl) (**I**), (**L2**)₂Pd(acyl)(Cl) (**II**), (**L1**)Pd(aryl)(Cl) (**III**), and (**L2**)₂Pd(aryl)(Cl) (**IV**) as potential intermediates. Thus, we synthesized and characterized them independently. Analysis of the model reaction under catalytically relevant conditions by ³¹P NMR revealed the detectable presence of five Pd complexes (Figure 3A), including

A. In situ ^{31}P NMR spectrum of catalytic reaction

B. Kinetics of decarbonylation from Pd(acyl) complexes



C. Kinetics of product formation from Pd(aryl) complexes



D. Proposed catalytic cycle

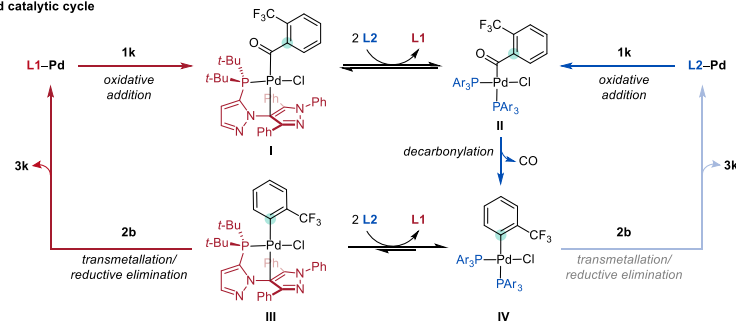


Figure 3. (A) Reaction monitoring by ^{31}P NMR. (B) Stoichiometric studies of the decarbonylation processes from complexes I and II. Yields of III and IV were determined by quantitative ^{31}P NMR. Light grey lines are for visual reference. (C) Stoichiometric studies of transmetalation/reductive elimination processes from complexes III and IV. Yield of 3k was determined by quantitative ^{19}F NMR relative to an internal standard. (D) Proposed catalytic cycle.

complexes I, II, and IV, but not III. While that observation provides an initial indication that III might not participate in the cooperative mechanism, we nonetheless proceeded to evaluate the kinetic competence of each of the four complexes in the relevant decarbonylation and transmetalation (TM)/reductive elimination (RE) steps through stoichiometric rate studies.

Second, the decarbonylation step was investigated with complexes I and II. The two complexes were found to decompose at drastically different rates (Figure 3B). At 70 °C, complex II underwent quantitative decarbonylation to complex IV within 5 min, whereas complex I remained stable over 2 h under the same conditions (Figure 3B, left). At 50 °C, the rate

of decarbonylation for complex II could be monitored conveniently and corresponds to a Gibbs free energy of activation $\Delta G^\ddagger = 23.5 \text{ kcal mol}^{-1}$. The large rate difference is consistent with the vacant coordination site required for decarbonylation/ α -elimination being kinetically more accessible with L2 than with the chelating L1 ligand.⁴³ It is further consistent with the relative barriers computed by neural-network potential (NNP) and density-functional theory (DFT),⁴⁴ where $\Delta\Delta G^\ddagger_{\text{comput}} = 2.8 \text{ kcal mol}^{-1}$ favors reaction from complex II (Figure 4). Furthermore, ligand exchange between I and free L2 was deduced to be facile at 70 °C, as the exposure of I to L2 leads to rapid conversion to L2-supported complexes II and IV. Taken together, these findings point

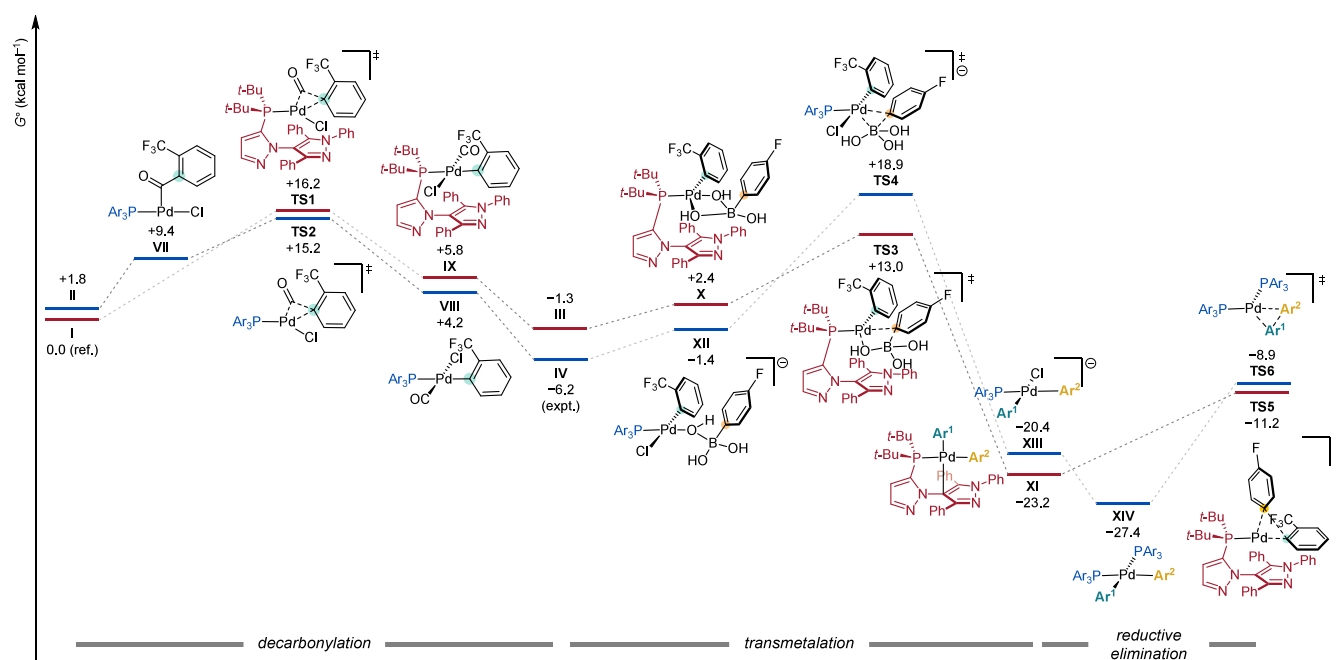


Figure 4. Computed free-energy profile for Pd-catalyzed decarbonylative cross-coupling for **L1** and **L2**. All calculations were performed at the ω B97M-V/def2-TZVPP/SDD(Pd)/SMD(MeCN) // UMA-M level of theory at 343.15 K and 0.75 mM for all species except **2b** (187.5 mM). See SI Table S16 for comparisons to full DFT calculations and Figures S15 and S16 for consideration of alternative pathways to decarbonylation, TM, and RE.

toward complex **I** being kinetically incompetent. Thus, decarbonylation proceeds only from **L2** complex **II** in the full dual-ligand catalytic system.

Third, the relative rates of stoichiometric TM and RE from complexes **III** and **IV** were investigated by treating each complex with boronic acid and base and monitoring formation of product **3k** (Figure 3C). Complex **III** exhibits a higher initial rate of product formation than **IV** (4.0 mM s^{-1} vs 2.2 mM s^{-1}) but more rapid deactivation (0.57 mM s^{-1} vs 1.5 mM s^{-1} at ca. 2 h) (Figure 3C). The kinetic behavior of **III** reflects productive TM/RE competing with a relatively rapid decomposition pathway that does not generate product **3k**. That observation is consistent with observed Pd black formation and with the known thermal instability of **L1**Pd-(aryl)(Cl) complexes in solution.^{45,46} Crucially, product formation from complex **IV** in the presence of added **L1** exhibits the superior initial rate of **III** (4.4 mM s^{-1}) with more sustained reaction rates than either **III** or **IV** alone (2.7 mM s^{-1} at ca. 2 h) (Figure 3C, left, purple trace). In separate experiments, ligand exchange ($\text{III} + 2 \text{ L2} \rightleftharpoons \text{IV} + \text{L1}$) was established to be facile, with **L2**-supported complex **IV** being favored thermodynamically by $4.9 \text{ kcal mol}^{-1}$ (Figure 4). We propose that these observations suggest a basis for ligand cooperativity, wherein **IV** serves as a stable resting state and a reservoir for the generation of a low steady-state concentration of more highly reactive **III**, while simultaneously protecting **III** from decomposition that occurs potentially through higher order pathways. This interpretation comports with the fact that **III** is not observed in the *in situ* ^{31}P NMR spectrum (Figure 3A).

For computational modeling of the TM step, we noted that the potential mechanistic pathways may be numerous and complex.^{47–49} Drawing from prior experimental and computational studies,^{30,48} we considered both anionic and neutral pathways involving **L1**, as well as base-promoted, outer-sphere

transmetalation for **L2**. Among the pathways considered computationally (Figures S15, S16), the lowest energy barriers for the three scenarios (Figure 4) align with experimental observations: **III** ($14.3 \text{ kcal mol}^{-1}$), **IV** ($25.0 \text{ kcal mol}^{-1}$), and **IV** + **L1** ($19.2 = 14.3 + 4.9 \text{ kcal mol}^{-1}$). Essentially, adding **L1** to **IV** forms **III** in low concentrations, but **III** forms product so rapidly that the net effect is a measurable increase in product formation.

A catalytic cycle consistent with all available mechanistic and computational data is provided in Figure 3D. The key feature is that the cooperative interplay of ligands **L1** and **L2** arises from dynamic ligand-relay catalysis:⁴² **L2** provides a lower-barrier pathway to decarbonylation and plays a protective role in the TM/RE step, **L1** provides a lower-barrier pathway to TM/RE, and facile **L1**/**L2** ligand exchange enables the system to access the more competent species for each elementary step.

CONCLUSION

By exploiting ligand cooperativity, we have accomplished decarbonylative Suzuki–Miyaura cross-couplings between aryl chlorides and boronic acids under mild conditions with broad compatibility across arene and heteroarene substrates. Key to the generality and efficiency was the combined use of BippyPhos **L1** and triarylphosphine **L2**, a ligand pair identified by agnostic and empirical screening for cooperativity through pooling–deconvolution.⁴⁰ Beyond validating the pooling–deconvolution algorithm, this study provides compelling evidence for the potential viability of multiligand approaches to broadly enhance (de)insertive metal-catalyzed cross-coupling reactions through ligand specialization.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.5c08167>.

Experimental synthetic procedures and characterization data for products and Pd complexes, procedures and details of mechanistic studies, and data from computational studies (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. M.H.S. and Y.J. contributed equally.

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Notes

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